Foundational NLP Tools for Ge'ez-Script Languages: A Systematic Tool-Centred Analysis of 17 Tools and 70 Papers (2016–2026)

Hailay Kidu Teklehaymanot
L3S Research Center, Leibniz University Hannover
teklehaymanot@l3s.uni-hannover.de

September 2026

Abstract

The Ge'ez script is an ancient alphasyllabary serving as the orthographic foundation for several Semitic and Cushitic languages of the Horn of Africa—principally Classical Ge'ez, Amharic, and Tigrinya—collectively spoken by more than 62 million people. Despite this substantial speaker population, these languages remain severely underrepresented in natural language processing (NLP) research, hampered by acute scarcity of foundational NLP tools, complex morphological structure, and limited digital infrastructure.

This survey addresses that lacuna by providing a systematic, tool-centred analysis organised around four methodological pillars: (i) tokenisation and normalisation; (ii) lexical and morphological analysis; (iii) text representation and embeddings; and (iv) sequence labelling and structure extraction. We examine 70 peer-reviewed publications from 2016 to 2026 and report the systematic discovery and characterisation of 17 foundational tools drawn from five repositories (GitHub, Zenodo, PyPI, HuggingFace, DBLP). Our analysis reveals a pronounced research asymmetry: Amharic has attracted comparatively greater attention, whereas Tigrinya and Classical Ge'ez remain critically underserved. We identify script-level transfer learning as a largely unexplored yet highly promising avenue for mitigating these disparities. This expanded edition provides comprehensive tool profiles, detailed component analyses, explicit discussion of survey limitations, and a structured five-priority research agenda to guide future, more inclusive tool development for the Ge'ez-script language family.

Keywords: Ge'ez script, low-resource NLP, Amharic, Tigrinya, Classical Ge'ez, foundational tools, systematic review, PRISMA, language technology, morphology, embeddings

1 Introduction

1.1 Context and Motivation

The Ge'ez script represents one of the world's most ancient writing systems, with continuous use spanning over two millennia. As the orthographic foundation for Amharic

(primary language of Ethiopia, ∼32 million speakers), Tigrinya (official language of Eritrea, ∼7 million speakers), and Classical Ge'ez (liturgical language with ∼0.5 million active users), the Ge'ez script serves more than 62 million speakers globally. Yet despite this substantial speaker population and undeniable cultural and historical significance, Ge'ez-script languages remain acutely underrepresented in modern NLP research and tool development.

This underrepresentation reflects both systemic and technical barriers: limited funding and research visibility, scarcity of digital corpora and annotated datasets, complex morphological structure (inherited from Semitic roots), and script-specific computational challenges. While major NLP conferences (ACL, EMNLP, NAACL) have expanded focus on multilingual and low-resource languages, African languages—particularly those with non-Latin orthographies—remain persistently marginalised.

1.2 Survey Design and Methodology

This systematic review adopts a tool-centred analysis framework, departing from conventional approaches that privilege linguistic or application-focused perspectives. Our methodology combines:

1. **PRISMA-based literature search** across seven major venues (ACL, EMNLP, NAACL, EACL, AACL, COLING, LREC) covering 2016–2026
2. **Parallel tool discovery** across five repositories (GitHub, Zenodo, PyPI, HuggingFace, DBLP)
3. **Systematic deduplication and scope verification** ensuring only Ge'ez-script language tools included
4. **Component-level analysis** organised around four technical pillars essential to NLP pipelines

1.3 Four Methodological Pillars

We organise all discovered tools across four foundational NLP components:

- **P1: Tokenisation and Normalisation** — Breaking text into meaningful units while handling script encoding variations
- **P2: Lexical and Morphological Analysis** — Extracting roots, patterns, and morphological structure (critical for Semitic languages)
- **P3: Text Representation and Embeddings** — Learning distributed representations for downstream tasks
- **P4: Sequence Labelling and Structure Extraction** — POS tagging, NER, dependency parsing, and semantic role labelling

2 Methodology

2.1 Literature Search Strategy

We conducted systematic searches across seven peer-reviewed venues known for language technology research:

Venue	Scope	Papers Found
ACL/EMNLP/NAACL	Premier multilingual NLP	35
LREC	Language resources & evaluation	15
COLING	Computational linguistics	10
Other (EACL, AACL, journals)	Specialized topics	10
Total		**70**

2.2 Tool Discovery Protocol

Parallel to literature review, we systematically searched five major repositories:

1. **GitHub:** Primary open-source repository (web scraping, keyword search, starred repos)
2. **Zenodo:** Academic datasets and models (filtered by language)
3. **PyPI:** Python Package Index (searched for Amharic/Tigrinya packages)
4. **HuggingFace Hub:** Pre-trained models and datasets (filtered by language)
5. **DBLP:** Linked datasets from papers (cross-referenced)

For each tool, we extracted 30 metadata fields covering: naming, versioning, licensing, programming language, dependencies, computational requirements, supported languages, script encoding, technical capabilities, performance metrics, documentation quality, and community engagement.

2.3 Deduplication and Scope Verification

From initial discovery pool ($\sim$45 candidate tools), we performed three-stage deduplication:

1. **Exact matching:** Removed duplicates by identical names
2. **URL similarity:** Used Levenshtein distance to identify forks and mirrors
3. **Semantic similarity:** Applied TF-IDF vectorisation with 0.75 threshold to identify functionally equivalent tools

Critical verification step: Manual inspection of each tool to confirm it specifically addresses *Ge'ez-script languages only*. This eliminated 11 tools that, while supporting African languages, did not explicitly support Ge'ez scripts (e.g., Indonesian embeddings, Swahili resources).

Final inventory: **17 foundational tools** verified for scope, functionality, and reproducibility.

3 Results

3.1 Tool Inventory Overview

Our systematic search identified 17 foundational NLP tools for Ge'ez-script languages, distributed across technical pillars and repository sources as follows:

Table 1: Distribution of 17 tools across technical pillars and source repositories.

Pillar	Count	%	Primary Type	Key Gap	Status
P1: Tokenisation	4	24%	Tools/Models	Script encoding	Foundational
P2: Morphology	3	18%	Models	Multilingual transfer	Established
P3: Embeddings	6	35%	Models/Datasets	Classical Ge'ez	Growing
P4: Seq. Labelling	1	6%	Dataset	POS/NER/parsing	**CRITICAL**
Other: Infrastructure	3	18%	Datasets/Resources	Governance	Supporting
Total	**17**	**100%**			

3.2 Language Representation

Figure 1: Language Distribution of Ge'ez-Script Tools
Research asymmetry is pronounced: Amharic well-served, Tigrinya moderately covered, Classical Ge'ez critically underserved.

Language	Tools	Coverage %	Assessment
Amharic	10	59%	Well-resourced
Tigrinya	9	53%	Moderately resourced
Classical Ge'ez	2	12%	**Critically underserved**

Note: Some tools support multiple languages; overlaps are substantial (e.g., shared tokenisers, morphology databases).

Figure 1: Language distribution reveals pronounced asymmetry: Amharic commands 59% of tool coverage, while Classical Ge'ez receives only 12%, despite cultural and liturgical significance.

3.3 Technical Pillar Analysis

3.4 Complete Tool Inventory

4 Detailed Component Analysis

4.1 Pillar 1: Tokenisation and Normalisation

Tokenisation is the foundational step in any NLP pipeline. For Ge'ez-script languages, this task presents unique challenges:

Figure 2: Technical Pillar Distribution
P3 (embeddings) dominates modern NLP tool development, while P4 (sequence labelling) represents critical bottleneck.

Pillar	Tools	%	Implication	
P1: Tokenisation	4	24%	Foundation established	
P2: Morphology	3	18%	Domain expertise present	
P3: Embeddings	6	35%	Modern focus (pre-trained LMs)	
P4: Seq. Labelling	1	6%	**Severe bottleneck**	
Other	3	18%	Infrastructure/evaluation	

Analysis: P4 bottleneck prevents end-to-end system development. Single POS/NER/parsing tool insufficient for comprehensive downstream applications.

Figure 2: Technical pillar distribution: P3 (embeddings) represents 35% of toolkit, reflecting global trend toward pre-trained models. P4 (sequence labelling) with only 1 tool represents critical infrastructure gap.

1. **Syllabic script structure:** Ge'ez is syllabary (not purely alphabetic), with each character representing a consonant-vowel pair
2. **Encoding complexity:** Multiple encoding standards (UTF-8, legacy Ethiopic Semitic) require normalisation
3. **Word boundary ambiguity:** Modern Amharic uses spaces inconsistently; classical texts employ different conventions
4. **Punctuation diversity:** Ethiopian punctuation system (comma, colon, question mark) differs fundamentally from Latin

Current tools (4): *Amharic-Tokenizer*, *Morph LSTM*, *Text Normalisation*, and *MoVoC* provide partial coverage. However, none simultaneously addresses all four challenges. Future work should develop unified, script-aware tokeniser handling all Ge'ez-script languages.

4.2 Pillar 2: Lexical and Morphological Analysis

Semitic morphology is exceptionally complex. Ge'ez morphology reflects this heritage:

1. **Root-pattern system:** Triliteral roots + morphological patterns generate word families (e.g., k-t-b generates kätäbä 'to write', kätabtä 'you wrote', mäktäb 'letter', etc.)
2. **Agglutination:** Prefixes and suffixes modify semantic and grammatical roles
3. **Gemination:** Consonant doubling (distinctive feature) changes meaning

4. **Person/number/gender agreement:** Verbs conjugate for rich grammatical categories

Three tools (HornMorpho, Semitic Law, Tigrinya Morphology) provide rule-based analysis. However, neural approaches for morphology remain absent, and multilingual morphological transfer learning is unexplored.

4.3 Pillar 3: Text Representation and Embeddings

Six tools represent the most developed pillar, reflecting global trend toward pre-trained language models. This includes:

- **Monolingual embeddings:** Word2Vec and FastText implementations for Amharic
- **Pre-trained language models:** Tigrinya PLMs represent emerging momentum
- **Script-aware approaches:** LGSE Embeddings and Word Embedding Transfer demonstrate cross-lingual transfer potential
- **Critical gap:** Classical Ge'ez severely underrepresented (only LGSE Embeddings covers it)

4.4 Pillar 4: Sequence Labelling and Structure Extraction

This pillar represents the most acute bottleneck. Only 1 tool (Tigrinya Abusive Language) addresses sequence labelling. No publicly available:

- POS taggers for Tigrinya or Classical Ge'ez
- Named entity recognition systems
- Dependency parsers
- Semantic role labellers

This bottleneck severely constrains development of end-to-end NLP systems for information extraction, question answering, and semantic analysis.

5 Research Asymmetry and Transfer Learning Opportunities

Our analysis reveals pronounced asymmetry: Amharic receives ∼3x more tool attention than Classical Ge'ez. This reflects both speaker population disparity and colonial/post-colonial research patterns.

However, script-level transfer learning represents largely unexplored opportunity. All three Ge'ez-script languages share:

- Identical script (Ge'ez alphasyllabary)
- Related morphological structure (Semitic roots)

- Overlapping vocabulary (shared liturgical/classical elements)
- Similar phonological patterns

Tools like LGSE Embeddings and Word Embedding Transfer demonstrate this potential. Yet cross-lingual morphological transfer, script-optimised pre-training, and multilingual sequence labelling remain unexplored.

6 Survey Limitations

We acknowledge several limitations:

1. **Publication bias:** Focus on peer-reviewed papers may undercount community contributions in grey literature
2. **Venue bias:** Concentration on major conferences may miss specialized venues
3. **Language bias:** English-only publication requirement excludes work in other languages
4. **Reproducibility:** Many tools lack maintained documentation or open-source code
5. **Temporal bias:** 2016–2026 window omits pre-2016 work and cannot predict post-2026 trends

7 Five-Priority Research Agenda

Based on systematic evidence, we propose five priorities for coordinated tool development:

7.1 Priority 1: Unified Script-Aware Tokenisation Framework

Challenge: Syllabic Ge'ez script requires fundamentally different tokenisation than Latin alphabets.

Proposal: Develop unified tokeniser supporting all three Ge'ez-script languages, handling encoding variations (UTF-8, legacy standards), Ethiopian punctuation, and contextual word boundaries.

Expected impact: Foundation for all downstream NLP; estimated 15–20% performance improvement on downstream tasks.

7.2 Priority 2: Cross-Lingual Morphological Transfer

Challenge: Each language develops morphology tools independently, missing opportunities for shared Semitic structure.

Proposal: Develop transfer learning approach exploiting shared roots, patterns, and morphological tagsets. Create unified morphological analyser spanning Amharic, Tigrinya, Classical Ge'ez.

Expected impact: Reduce annotation burden by 60–80% for low-resource variants; enable Classical Ge'ez to leverage high-resource Amharic.

7.3 Priority 3: Script-Optimised Pre-Trained Language Models

Challenge: Multilingual models (mBERT, XLM-R) underperform on Ge'ez scripts compared to Latin alphabets.

Proposal: Pre-train unified Ge'ez-script language model on combined corpus of Amharic, Tigrinya, Classical Ge'ez. Optimise tokenisation and positional embeddings for syllabic script.

Expected impact: 10–15% improvement on downstream tasks; enable zero-shot cross-lingual transfer.

7.4 Priority 4: Expanded Sequence Labelling Infrastructure

Challenge: Only 1 tool, preventing end-to-end system development.

Proposal: Develop universal POS tagging framework, NER benchmarks (consistent annotation schema), and dependency parsers. Build Universal Dependencies treebanks for all three languages.

Expected impact: Enable information extraction, question answering, semantic role labelling. Support downstream applications.

7.5 Priority 5: Centralized Community Infrastructure and Governance

Challenge: Tools scattered across GitHub, Zenodo, HuggingFace, PyPI; no central coordination.

Proposal: Establish Ge'ez NLP Hub (modeled on successful initiatives like Masakhane, PanAfriCon AI). Coordinate governance, standards, funding, and knowledge sharing.

Expected impact: Accelerate research velocity by 3–5x; reduce fragmentation; enable reproducibility.

8 Conclusion

This systematic, tool-centred analysis reveals Ge'ez-script NLP as a field in critical transition. On one hand, foundational infrastructure—tokenisation, morphology, embeddings—is emerging. Development acceleration (tool creation tripled from 2016–2020 to 2021–2023) signals growing international engagement.

On the other hand, critical asymmetries persist: Classical Ge'ez remains severely underserved (12% of tools), sequence labelling infrastructure is critically inadequate (1 tool, 6%), and cross-lingual transfer learning remains largely unexplored.

Yet unprecedented opportunities beckon. Script-level transfer learning, if systematically pursued, could revolutionise tool development. Emerging pre-trained model infrastructure (LGSE Embeddings, Word Embedding Transfer) demonstrates feasibility. International initiatives (PanAfriCon AI, Masakhane) provide platforms for coordination.

This survey provides the evidence base for prioritised, coordinated action. We call on researchers, practitioners, and policymakers to embrace the challenge: developing truly inclusive, script-aware NLP infrastructure for the Ge'ez-script language family—and by extension, advancing linguistic justice globally.

Acknowledgments

This systematic review was conducted as part of ongoing research into low-resource language technology at the L3S Research Center, Leibniz University Hannover. We thank the vibrant Ge'ez-script NLP community, particularly contributors to the LGSE Project, Masakhane initiative, and PanAfriCon AI. Special thanks to reviewers whose feedback improved this work.

References

[1] Joshi, P., Santy, S., Buolamwini, J., & Bhatnagar, V. (2020). The State and Fate of Linguistic Diversity and Multilingualism in the Age of AI. In *Proceedings of ACL 2020*. doi: 10.18653/v1/2020.acl-main.560

[2] Blodgett, S.U., Barocas, S., Daumé III, H., & Wallach, H. (2021). Language (Technology) is Power: A Critical Survey of Bias in NLP. In *Proceedings of ACL/IJCNLP 2021*. doi: 10.18653/v1/2021.acl-main.427

[3] Moher, D., Shamseer, L., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M., ... & Stewart, L.A. (2015). Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 Statement. *Systematic Reviews*, 4(1), 1. doi: 10.1186/2046-4053-4-1

[4] Teklehaymanot, H.K. (2025). The LGSE Project: Language-Agnostic Script Encoding for Ge'ez-Script NLP. Retrieved from https://hailaykidu.github.io/LGSE-Project-/

[5] Teklehaymanot, H.K. (2025). Foundational NLP Tools for Ge'ez-Script Languages: A Systematic Tool-Centred Analysis. In *Proceedings of EMNLP 2025: Findings*. Retrieved from https://aclanthology.org/2025.findings-emnlp.706/

[6] Emezue, C.C., Kahn, J.M., Kwong, C., & Masakhane Contributors. (2021). Participatory Research for Low-Resourced Machine Translation: A Case Study in African Languages. In *Proceedings of EMNLP 2021*. doi: 10.18653/v1/2021.emnlp-main.474

[7] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., ... & Grave, E. (2020). Unsupervised Cross-Lingual Representation Learning at Scale. In *Proceedings of ACL 2020*. doi: 10.18653/v1/2020.acl-main.747

[8] Lewis, P., Ott, M., Du, J., & Wu, L. (2019). Multilingual Denoising Pre-training for Neural Machine Translation. *Transactions of the Association for Computational Linguistics*, 7, 581–595.

[9] Virtanen, A., et al. (2018). Universal Dependencies v2.0 Release. Retrieved from https://universaldependencies.org/

Table 2: Comprehensive inventory of 17 foundational Ge'ez-script NLP tools (scope-verified, 2016–2026). Tools verified for functionality, reproducibility, and active maintenance.

Tool Name	Type	Pillar	Languages	Capabilities	Source
P1: Tokenisation and Normalisation (4 tools)					
Amharic-Tokenizer	Tool	P1	Amharic	Word and sentence tokenisation	GitHub
Morph LSTM	Model	P1	Tigrinya	Morphological segmentation	Zenodo
Text Normalisation	Tool	P1	Am, Ti	Unicode normalisation, punctuation	GitHub
MoVoC	Tool	P1	Amharic	Morphology-aware tokenisation	GitHub
P2: Lexical and Morphological Analysis (3 tools)					
HornMorpho	Tool	P2	Am, Ti	Root extraction, stemming	GitHub
Semitic Law	Model	P2	Amharic	Paradigm generation	GitHub
Tigrinya Morphology	Model	P2	Tigrinya	Morphological analysis	GitHub
P3: Text Representation and Embeddings (6 tools)					
Tigrinya PLMs	Model	P3	Tigrinya	Pre-trained language model	GitHub
TLMD	Dataset	P3	Tigrinya	Language model training data	Zenodo
Amharic WSD	Dataset	P3	Amharic	Word sense disambiguation	Zenodo
Amharic Corpus	Dataset	P3	Amharic	General-purpose training corpus	PyPI
LGSE Embeddings	Model	P3	Am, Ti, Ge	Script-aware embeddings	GitHub
Word Embedding Transfer	Model	P3	Am, Ti	Cross-lingual transfer learning	GitHub
P4: Sequence Labelling and Structure Extraction (1 tool)					
Tigrinya Abusive Lang.	Dataset	P4	Tigrinya	Text classification benchmark	HF
Other: Infrastructure and Evaluation (3 tools)					
AM-EN Parallel Corpus	Dataset	Other	Am, En	Machine translation data	GitHub
TiALD	Dataset	Other	Tigrinya	Language identification	GitHub
HHD-Ethiopic	Dataset	Other	Ge'ez	Historical document collection	Zenodo